

Cramer's Rule

Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of  $A\vec{x} = \vec{b}$  is affected by changes in the entries of  $\vec{b}$ .

However, the formula is inefficient for hand calculations, except for  $2 \times 2$  or perhaps  $3 \times 3$  matrices.

For any  $n \times n$  matrix  $A$  and any  $\vec{b}$  in  $\mathbb{R}^n$ , let  $A_i(\vec{b})$  be the matrix obtained from  $A$  by replacing column  $i$  by the vector  $\vec{b}$ .

$$A_i(\vec{b}) = [\vec{a}_1 \dots \underset{\substack{\uparrow \\ \text{col } i}}{\vec{b}} \dots \vec{a}_n]$$

Theorem 7 - Cramer's Rule

Let  $A$  be an invertible  $n \times n$  matrix. For any  $\vec{b}$  in  $\mathbb{R}^n$ , the unique solution  $\vec{x}$  of  $A\vec{x} = \vec{b}$  has entries given by

$$x_i = \frac{\det A_i(\vec{b})}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

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Proof

Denote the columns of  $A$  by  $\vec{a}_1, \dots, \vec{a}_n$  and the columns of the  $n \times n$  identity matrix  $I$  by  $\vec{e}_1, \dots, \vec{e}_n$ . If  $A\vec{x} = \vec{b}$ , the definition of matrix multiplication shows that

$$\begin{aligned} A(I_i(\vec{x})) &= A[\vec{e}_1 \dots \vec{x} \dots \vec{e}_n] \\ &= [A\vec{e}_1 \dots A\vec{x} \dots A\vec{e}_n] \\ &= [\vec{a}_1 \dots \vec{b} \dots \vec{a}_n] = A_i(\vec{b}) \end{aligned}$$

By the multiplicative property of determinants,

$$(\det A)(\det I_i(\vec{x})) = \det A_i(\vec{b})$$

The second determinant on the left is simply  $x_i$ . (Make a cofactor expansion along the  $i$ th row). Hence,

$(\det A)x_i = \det A_i(\vec{b})$ . This proves (1) because  $A$  is invertible and  $\det A \neq 0$ .

### 3.3 Cramers rule, Volume and Linear Transformations

II

#### Example 1

Use Cramer's rule to solve the system

$$3x_1 - 2x_2 = 6$$

$$-5x_1 + 4x_2 = 8$$

#### Solution

View the system as  $A\vec{x} = \vec{b}$ . Using the notation introduced above,

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, \quad A_1(\vec{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(\vec{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since  $\det A = 2$ , the system has a unique solution.

[this follows from:

For a square matrix  $A$ , if  $\det A \neq 0$ , means the matrix is invertible. And if  $A$  is invertible, then  $A\vec{x} = \vec{b}$  has exactly one solution

By Cramer's rule,

$$x_1 = \frac{\det A_1(\vec{b})}{\det A} = \frac{24+16}{2} = 20$$

$$x_2 = \frac{\det A_2(\vec{b})}{\det A} = \frac{24+30}{2} = 27$$

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III

#### Application to Engineering

A number of important engineering problems, particularly in electrical engineering and control theory, can be analyzed by Laplace transforms. This approach converts an appropriate system of linear differential equations into a system of linear algebraic equations whose coefficients involve a parameter. The next example illustrates the type of algebraic system that may arise.

#### Example 2

Consider the following system in which  $s$  is an unspecified parameter. Determine the values of  $s$  for which the system has a unique solution, and use Cramer's rule to describe the solution.

$$\begin{aligned} 3sx_1 - 2x_2 &= 4 \\ -6x_1 + sx_2 &= 1 \end{aligned}$$

#### Solution

View the system as  $A\vec{x} = \vec{b}$ , then

$$\underbrace{\begin{aligned} 3sx_1 - 2x_2 &= 4 \\ -6x_1 + sx_2 &= 1 \end{aligned}}_{A\vec{x}} = \underbrace{\begin{aligned} 4 \\ 1 \end{aligned}}_{\vec{b}} \quad A = \begin{bmatrix} 3s & -2 \\ -6 & s \end{bmatrix}$$

$$A_1(\vec{b}) = \begin{bmatrix} 4 & -2 \\ 1 & s \end{bmatrix}, \quad A_2(\vec{b}) = \begin{bmatrix} 3s & 4 \\ -6 & 1 \end{bmatrix}$$

Computing  $\det A$

$$\det A = 3s^2 - 12 = 3(s+2)(s-2)$$

The system has a unique solution precisely when  $s \neq \pm 2$ . For such an  $s$ , the solution is  $(x_1, x_2)$ , where

$$x_1 = \frac{\det A_1(\vec{b})}{\det A} = \frac{4s + 2}{3(s+2)(s-2)}$$

$$x_2 = \frac{\det A_2(\vec{b})}{\det A} = \frac{3s + 24}{3(s+2)(s-2)} = \frac{\cancel{3}(s+8)}{\cancel{3}(s+2)(s-2)}$$

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IV

#### A formula for $A^{-1}$

Cramer's rule leads easily to a general formula for the inverse of an  $n \times n$  matrix  $A$ . The  $j$ th column of  $A^{-1}$  is a vector  $\vec{x}$  that satisfies

$$A\vec{x} = \vec{e}_j$$

where  $\vec{e}_j$  is the  $j$ th column of the identity matrix, and the  $i$ th entry of  $\vec{x}$  is the  $(i, j)$ -entry of  $A^{-1}$ . By Cramer's rule

$$\left\{ (i, j)\text{-entry of } A^{-1} \right\} = x_i = \frac{\det A_i(\vec{e}_j)}{\det A} \quad (2)$$

Recall that  $A_{ji}$  denotes the submatrix of  $A$  formed by deleting row  $j$  and column  $i$ . A cofactor expansion down column  $i$  of  $A_i(\vec{e}_j)$  shows that

$$\det A_i(\vec{e}_j) = (-1)^{i+j} \det A_{ji} = C_{ji} \quad (3)$$

where  $C_{ji}$  is a cofactor of  $A$ .

By (2), the  $(i, j)$ -entry of  $A^{-1}$  is the cofactor  $C_{ji}$  divided by  $\det A$ . [Note that the subscripts on  $C_{ji}$  are the reverse of  $(i, j)$ .] Thus

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \quad (4)$$

The matrix of cofactors on the right side of (4) is called the *adjugate* (or classical adjoint) of  $A$ , denoted  $\text{adj } A$ .

#### Theorem 8 An Inverse Formula

Let  $A$  be an invertible  $n \times n$  matrix. Then

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

### 3.3 Cramers rule, Volume and Linear Transformations

V

#### Example 3

Find the inverse of the matrix  $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$

#### Solution

Let's start by computing the nine cofactors of  $A$ .

(remember the sign order)

$$C_{11} = + \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2 \quad C_{12} = - \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3 \quad C_{13} = + \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$$

$$C_{21} = - \begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14 \quad C_{22} = + \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7 \quad C_{23} = - \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7$$

$$C_{31} = + \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4 \quad C_{32} = - \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1 \quad C_{33} = + \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

The adjugate matrix is the transpose of the cofactor matrix.

$$\text{adj } A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

We could compute  $\det A$  directly, but the following computation provides a check on the calculations for  $\text{adj } A$  and produces  $\det A$ :

$$\text{adj } A \cdot A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{bmatrix} = 14I$$

Since  $(\text{adj } A)A = 14I$ , theorem 8 shows that  $\det A = 14$ .